Prefers-reduced-motion

In short

Moving parts on a website can be hard for people with ADHD, motion sickness, or balance problems. People can turn on a setting to reduce motion on their device. Use the prefers-reduced-motion CSS feature so motion only plays when they haven't asked for less.

What is it for?

Some people struggle with motion on websites. This can be due to a number of reasons, including (but not limited to):

- ADHD
- Motion sickness
- Vestibular disorders

While it's best not to use any motion that autoplays, sometimes it's unavoidable. To stop this creating barriers, use the prefers-reduced-motion media query. It stops the motion for people who have chosen to see less of it.

Reducing motion can have other benefits too, such as saving battery life and data on mobile devices.

Many new operating systems allow you to choose to reduce the motion that you see on websites (at least, on websites where they've used the prefers-reduced-motion media query).

If you're on a Mac, go to `System Settings`, then `Accessibility`, then `Display`, and turn `Reduce motion on or off`.

If you're on Windows, go to `Settings`, then `Accessibility`, then `Visual effects`, and turn `Animation effects on or off`.

See browser support for prefers-reduced-motion (Can I use).

How to support it

It's like any other media query (such as screen size). The two options available are `reduce` or `no-preference`. You can use it in CSS to turn off an element's motion.

If you're providing moving media to developers, also provide a non-moving variant that they can use in the media query.

If you're a developer, there are two ways you can use it. Either you can reduce the motion if the user has set a preference to reduce motion, like this:

```
.moving-element {
    animation: bounce 500ms;
}
@media (prefers-reduced-motion: reduce) {
    .moving-element {
        animation: none;
    }
}
```

Or you can set the motion to happen if the user hasn't set a preference. Here's an example of using it like this:

```
@media (prefers-reduced-motion: no-preference) {
    .moving-element {
        animation: bounce 500ms;
    }
}
```

This is the recommended way, because browsers that don't support the prefers-reduced-motion media query will ignore this code, so your element will have no motion as standard. It's also less code. People who haven't turned on the setting will still see the motion, in the same way as the first method.

Useful links

- Web Content Accessibility Guidelines (WCAG) 2.2 Techniques - Using the CSS prefers-reduced-motion query to prevent motion
- Respecting users' motion preferences
- MDN - prefers-reduced-motion

Abbreviations

The full form of each abbreviation used in this document.

ADHD: Attention deficit hyperactivity disorder

CSS: Cascading Style Sheets

MDN: Mozilla Developer Network

WCAG: Web Content Accessibility Guidelines